

Low-Rank Tensor Approximation over $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$

Donghyun Kim Beomjun Chung Suho Cho Sabin Park
TP406 — Geometry of Tensors and Applications — Team 4 — 2026

1. Problem Statement

Find the best **rank-2** approximation of any tensor $T \in \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$:

$$T' = \arg \min_{\hat{T} \in \sigma_2^0(\text{Seg}(\mathbb{C}^3 \times \mathbb{C}^3 \times \mathbb{C}^3))} \|T - \hat{T}\|_F$$

where σ_2^0 is the set of *strict* rank-2 tensors:

$$T' = \mathbf{a}_1 \otimes \mathbf{b}_1 \otimes \mathbf{c}_1 + \mathbf{a}_2 \otimes \mathbf{b}_2 \otimes \mathbf{c}_2, \quad \mathbf{a}_i, \mathbf{b}_i, \mathbf{c}_i \in \mathbb{C}^3$$

Segre variety. $\text{Seg}(\mathbb{C}^3 \times \mathbb{C}^3 \times \mathbb{C}^3)$ is the variety of rank-1 tensors. Its r -th secant variety σ_r is the closure of the set of tensors of rank $\leq r$. We seek $T' \in \sigma_2^0$, the open dense part of σ_2 .

2. Distance: Frobenius Norm

Definition (entry-wise generalization to tensors):

$$\|T\|_F = \sqrt{\sum_{i,j,k=1}^3 |T_{ijk}|^2} = \sqrt{\langle T, T \rangle}$$

Why Frobenius?

CP decomposition algorithms are *derived* from this norm — ALS subproblems become standard least-squares problems

Unitary invariance: $\|T \times_n Q_n\|_F = \|T\|_F$ for unitary Q_n

$\|\cdot\|_F^2$ is smooth $\Rightarrow$ clean gradient derivations

Relative error used in all tests:

$$\varepsilon_{\text{rel}} = \frac{\|T - T'\|_F}{\|T\|_F} \times 100\%$$

Numerator $\|T - T'\|_F$: residual (how wrong)

Denominator $\|T\|_F$: scale of original tensor

3. Geometric Interpretation

Ambient space: $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3 \cong \mathbb{C}^{27}$.

Rank-2 variety (expected affine dimension):

$$\dim \sigma_2 \approx 2 \times \frac{(3+3+3-2)}{\dim \text{Seg}} + 1 = 15 \quad (\text{proj.}) \Rightarrow 16 \quad (\text{affine})$$

Full space: $\mathbb{C}^{27}$ (27 dim)

$\sigma_2 \approx 16$ dim

residual ≈ 11 dim

A random tensor in $\mathbb{C}^{27}$ has ≈ 11 dimensions orthogonal to σ_2 — directly explaining the 40–50% error observed for generic tensors.

4. CP Decomposition

Represent T as a sum of rank-1 tensors:

$$T' = \sum_{r=1}^R \mathbf{a}_r \otimes \mathbf{b}_r \otimes \mathbf{c}_r$$

Factor matrices:

$$A = [\mathbf{a}_1 | \dots | \mathbf{a}_R] \in \mathbb{C}^{3 \times R}, \quad B \in \mathbb{C}^{3 \times R}, \quad C \in \mathbb{C}^{3 \times R}$$

Mode- n unfolding $T_{(n)}$ rearranges T into a matrix (mode- n index $\rightarrow$ rows). The CP structure reads:

$$T_{(0)} = A(C \odot B)^\top$$

$$T_{(1)} = B(C \odot A)^\top$$

$$T_{(2)} = C(B \odot A)^\top$$

where $\odot$ is the **Khatri–Rao product** (column-wise Kronecker): $(A \odot B)_r = \mathbf{a}_r \otimes \mathbf{b}_r$.

5. CP-ALS Algorithm

Objective: minimise $\|T - T'\|_F^2$ over A, B, C .

Each mode's subproblem (others fixed) is a **least-squares** problem. Normal equations give the closed-form update:

$$A \leftarrow T_{(0)} \overline{K_{BC}} \overline{G}^{-1}$$

$$B \leftarrow T_{(1)} \overline{K_{AC}} \overline{G}^{-1}$$

$$C \leftarrow T_{(2)} \overline{K_{AB}} \overline{G}^{-1}$$

where:

$$K_{BC} = C \odot B \quad (\text{Khatri–Rao product})$$

$$G = (A^\dagger A) \otimes (B^\dagger B) \otimes (C^\dagger C) \quad (\text{Hadamard product of Gram matrices})$$

$\overline{(\cdot)}$ = complex conjugate (for real T : reduces to standard ALS)

Convergence criterion:

1 iteration = $A \rightarrow B \rightarrow C$ (one full cycle).

After each iteration, compute $\varepsilon_k = \|T - T'_k\|_F / \|T\|_F$.

Stop if $|\varepsilon_{k-1} - \varepsilon_k| < \tau$ ($\tau = 10^{-8}$, default).

Hard cap at $k_{\max} = 2000$ iterations.

Repeat $n_{\text{restart}} = 10$ times with different random initialisations; keep the result with smallest ε .

6. Implementation

Platform: Python 3.13 · NumPy only (no TensorLy dependency)

File	Role
cp_als.py	CP-ALS core; real & complex tensors
main.py	4-case demo with factor printout
test_suite.py	3-category test with statistics

Usage:

```
from cp_als import rank2_approximation
T_prime, factors, abs_err, rel_err = \
    rank2_approximation(T, n_restarts=10)
A, B, C = factors # each shape (3, 2)
```

```
$ python test_suite.py --trials 10
$ python test_suite.py --complex
$ python test_suite.py --noise-levels 0.01 0.1 1.0
```

7. Test 1: Exact Rank-2 (Sanity Check)

T is an exact rank-2 tensor; ALS must recover it to near-zero error.

	Pass rate	Mean ε_{rel}	Max ε_{rel}
Real	10/10	$\sim 10^{-6}\%$	$\sim 10^{-5}\%$
Complex	10/10	$\sim 10^{-5}\%$	$\sim 10^{-4}\%$

Conclusion: ALS recovers exact rank-2 tensors to floating-point precision in all 10/10 trials.

8. Test 2: Noisy Rank-2 (Stability)

$T = T_{\text{clean}} + \varepsilon \cdot N$, $\|N\|_F = 1$, T_{clean} is exact rank-2.

Noise ε	Mean ε_{rel} (real)	vs. noise level
1.00%	0.19%	$\ll \varepsilon$
5.00%	0.95%	$< \varepsilon$
10.00%	1.89%	$< \varepsilon$
30.00%	5.67%	$< \varepsilon$
50.00%	9.30%	$< \varepsilon$
100.00%	17.0%	$< \varepsilon$

Conclusion: Error is always *smaller* than the noise level and scales proportionally — the algorithm is numerically stable near the rank-2 manifold.

9. Test 3: Generic Random Tensor

$T \sim \mathcal{N}(0, 1)^{3 \times 3 \times 3}$, no structure assumed.

	Real	Complex
Mean ε_{rel}	45.5%	50.3%
Range	31.8–58.9%	40.2–59.1%

Why 40–50%? σ_2 occupies ≈ 16 of 27 dimensions. A random tensor has ≈ 11 components outside σ_2 that cannot be captured by any rank-2 approximation. This is a *geometric limitation*, not an algorithmic failure.

10. Conclusions

Correctness: CP-ALS with Frobenius norm recovers exact rank-2 tensors to machine precision.

Stability: Approximation error scales linearly with noise and remains below the noise level.

Generic case: 40–50% residual for random tensors is a geometric consequence of $\dim \sigma_2 \approx 16 < 27$.

Norm choice: Frobenius norm yields closed-form ALS updates and unitary invariance, consistent with the algebraic geometry framework of the problem.

11. Future Work

Compare ALS with gradient-based methods (Adam, L-BFGS)

Investigate border rank: when does $T' \in \sigma_2^0$ vs. only in σ_2 ?

Quantify rank–quality tradeoff empirically (rank 3, 4)

Riemannian optimisation directly on σ_2^0